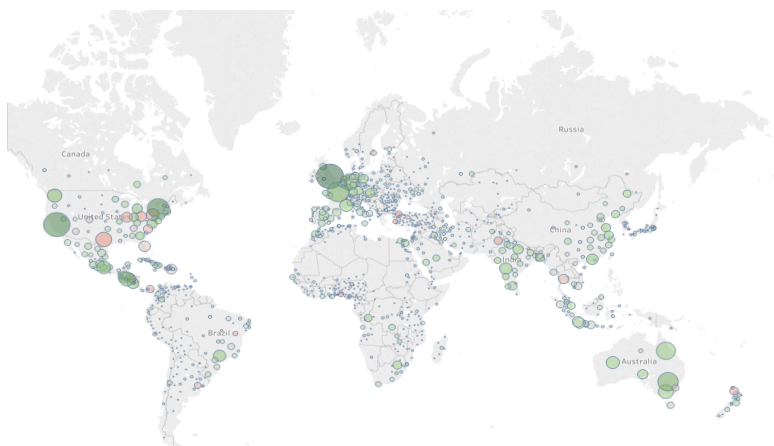




DATA VISUALIZATION WORKSHOP

I320U INFORMATION & INTERACTION DESIGN

DR. MCQUAID

INTRO

Go through a Getting Started exercise with Tableau. This comes from the Tableau website, so you can always go back over this by visiting the Learning portion of the site.

GLOBAL SUPERSTORE

We'll use an Excel spreadsheet called Global Superstore, shaped as a table recording orders, returns, and people for a fictional store operating worldwide. This is *not* the Sample Superstore spreadsheet! You will quickly get lost if you open the Sample Superstore spreadsheet!

The Global Superstore spreadsheet is in Canvas > Files > visualization > Global Superstore.xls so start by downloading it.

DOWNLOAD TABLEAU

Next, download a trial copy of Tableau. Students can obtain a special license for free but don't worry about that right now. The trial copy will serve our purposes.

OPEN THE APPLICATION

- The first thing you should see is a screen with headings including Connect and Open and Discover.
- Look at More Servers briefly. This shows the range of data sources you can use.
- Return to the main screen.

OPEN THE GLOBAL SUPERSTORE SPREADSHEET

- Choose Excel from the Connect pane. Double-click on the Global Superstore spreadsheet to select it.
- Now you're in the Data Connection window. From here, you can see that there are three worksheets in the Excel file, Orders, People, and Returns.

CHOOSE THE ORDERS WORKSHEET

- Drag Orders to the Canvas, which is the large white rectangle. You will see the first few rows of that worksheet, which is organized as a table with each row representing information about a particular order and each column representing an attribute. This is the way a database table would be laid out.
- You can now join this table to one of the other tables by dragging Returns to the canvas. This will reveal a path between the two tables. You can select a way to join these tables together using the dialog at the bottom of the screen.

- Throughout the rest of this tutorial, though, we are going to only use Orders. Click Remove from the options triangle on the Returns icon.
- Look at the preview of the data in the lower part of the screen. You can click on each name to rename it. Click on the icon # beside the Row ID and change the data type from integer (whole number) to string.
- Another thing you can do is to break up a piece of data into its parts. The Order ID, for instance, includes a two letter code for the distribution center as well as four digits for the year and an integer for the productid. Create a separate column for distribution center with the following steps. (1) Click on the options triangle and select split. (2) Delete the Split 2 column and the Split 3 column. (3) Rename the Split 1 column as Distribution Center. (Note that the split columns show up at the end of the list of columns!)
- In the upper right screen corner, you can choose whether the connection to the spreadsheet is live or if, instead, you extract the data in the spreadsheet into Tableau. Leave it live for now but be aware that, if you are connected to an enterprise database, you may reduce server load by extracting data.

GO TO THE TABLEAU WORKSHEET

- Click on the Sheet 1 tab at the bottom of the screen. This is where you will do the main work.
- Look at the parts of the screen. There is a ribbon at the top, a data pane on the left, the main workspace in the center, and a show me pane on the right. Above the main workspace and below the ribbon are two shelves: Columns and Rows.
- The data pane contains two kinds of data: categorical and numeric. The numeric items are called *Measures*. Earlier versions of Tableau called the categorical items *Dimensions*

so you may see that term from time to time. The two parts of the data pane are separated by a horizontal line between the categorical items and the numeric items.

CREATE A DATA VISUALIZATION

- Drag the word Category from the Dimensions part of the data pane to the Rows shelf. You should see three rows representing the categories in Sheet 1.
- Drag Quantity from the Measures part of the data pane to the Columns shelf. You should see sideways barchart of of Quantity by Category in Sheet 1.
- Drag Segment from Dimensions to Rows. The barchart expands with three segments per category: Consumer, Corporate, and Home Office.
- Drag Market from Dimensions to Columns. The barchart expands into seven smaller barcharts, one for each market.
- Drag Market again, this time from Dimensions to Color in the Marks shelf. Now each barchart is colored according to its market.

LOOK AROUND

The items you dragged into the Columns shelf and Rows shelf are called Pills in Tableau parlance. Notice that the Measures pill, SUM(Quantity), is colored green while the Dimensions pills are all colored blue. The measures pill represents a number in a more or less continuous distribution, while the dimensions are categorical, in more or less discrete distributions.

CLEAR SHEET

In the ribbon, select the Clear Sheet icon. It has a tiny barchart, partly covered by an ×.

A NEW VISUALIZATION

- Drag Sales from Measures into the left margin of Sheet 1. A single bar appears, showing the sum of sales for the Global Superstore to be about 12.6 million.
- Drag Order Date from Dimensions into Columns. The bar-chart auto-converts into a line chart showing sales over time and labeled by Year.
- Click the + symbol on the left of the YEAR(Order Date) pill. The line chart expands into four line charts, one for each year, and aggregated at the quarter level.
- Swap the positions of the two blue pills in Columns so that Quarter is in front of year. Now the view is changed so that each line chart shows one quarter over four years.
- Drag the YEAR(Order Date) pill out of Columns and onto Color on the Marks shelf. Now the years are in one stacked line chart.
- Click on the options triangle of the QUARTER(Order Date) pill. Click on Months to expand the stacked line chart according to months.
- Click on the options triangle of the SUM(Sales) pill. One of the choices there is Measure(Sum). Click there and you can see some other choices of Measures. Click on Average to see how that will look.
- Then click on Sum to return to the previous view. (Or click Undo.)
- Click on the options triangle of the SUM(Sales) pill again. This time, go to Quick Table Calculation and look at the submenu.
- Choose Year over Year Growth from the submenu. The line chart changes.
- Drag Sales from the Measures pane back into Rows and now you have two displays.
- Now move the SUM(Sales) that represents Year over Year Growth from Rows to the Tooltip in the Marks shelf. Now

the Year over Year Growth line chart disappears and, instead, year over year growth is shown as a tooltip when you hover over points in the remaining chart.

- Drag Category to the Rows shelf. Now the line chart is split into three charts, one for each category.
- The peak for office supplies in August reflects the annual back to school sale. Annotate the chart with that fact.
- Right click near that point and choose Annotate, Area from the resulting menu. Write in an annotation like “This is our annual back to school sale.” Click OK. If the annotation background is hiding part of the line, you can move it or decrease its shading on the format menu.
- Right click on the Sheet 1 tab at the bottom of the screen and rename this “Sales Seasonality.”

CREATE A SPREADSHEET FROM THE VISUALIZATION

Right click anywhere in the chart and choose Copy, Data from the resulting menu. If you open a copy of Excel you can paste it into a new workbook, where it will include the Year over Year growth calculation from the tooltips.

- Instead of exporting to Excel, you can view the data by going to the menu and choosing Worksheet > Duplicate as Crosstab
- Switch to the new tab, Sales Seasonality (2)
- Click on the Swap Rows and Columns icon in the ribbon so that months are rows.
- Move Category from the Columns shelf to the Rows shelf.
- Click on Standard on the ribbon and choose Entire View to make the crosstab take up the entire window space.
- Drag Profit from Measures to Color in the Marks shelf.
- Make the colors stand out more by clicking Color on the Marks shelf and choosing Edit Colors.

- Choose Red-Green Diverging, Use Full Color Range, and Stepped Color with 6 steps.
- Choose Undo on the ribbon and try a different way to emphasize profitability.
- Click the options triangle for the Marks shelf. It currently reads Automatic. Change it to Square. Now you just see colored squares.
- Click on Label on the Marks shelf and choose Show Mark Labels.
- You can change the colors back to red-green diverging if you wish, so it looks more like a traditional profit and loss display.
- Double click on the Sales Seasonality (2) tab and rename this Crosstab.

CREATE A SHEET TO EXPLORE GLOBAL SALES AND PROFITS

The Crosstab worksheet shows that Furniture is less successful than the other two categories in the Fall but it does not tell us where the problem lies. Use a new sheet to try to find the locations of furniture's profitability problems.

- Click on the New Worksheet icon next to the word Crosstab at the bottom of the display.
- Click on the Show Me tab if its window is not already visible. Drag the Show Me pane over near the Measures pane.
- Control-click (on Windows) or Command-click (on Mac) on several measures and see suggestions for appropriate charts highlight on the Show Me pane.
- Select Sales and Country (using Control-click if on Windows or Command-click if on Mac). One of the highlighted types is the symbol map. Select that. You should see a world map with dots of different sizes centered on Country territories. If not, drag Country and Sales to the map. They should appear as pills on the Marks shelf.

- Drag State onto the map and notice that the Sales circles are now broken down into smaller regions.
- Drag the Size slider on the Marks shelf to increase the size of the dots.
- Choose Color on the Marks shelf and adjust the opacity to around 50 percent.
- Also on Color add dark blue borders instead of leaving them set at Automatic.
- There should be a size legend beside the display. Click on it so the options triangle appears and select Hide Card on that menu.
- Drag Profit to Color. You may want to change the colors to Red-Green diverging.
- Search for United Arab Emirates using the magnifying glass on the map. The map should zoom into UAE. Select the pin icon to *unpin* the map, zooming out to the previous view.
- Drag Category to the Filter shelf. Choose Furniture in the dialog box. Now the map is restricted to showing profitability and sales of furniture. The color of the circles represents profit and the size of the circles represents sales.
- Right click the Category: Furniture pill and choose Show Filter on the resulting menu. Now anyone viewing the exported visualization can choose a category or categories of interest.
- Restrict the viewer's choice by clicking the options triangle on the Category filter. Currently Multiple Values (list) is checked. Change that by clicking Single Value (list). Now a viewer can only choose one category at a time.
- Rename this sheet Global Sales and Profits.

CREATE A SHEET TO EXPLORE SUB-CATEGORIES

Click on New worksheet.

Furniture is doing poorly but what types of furniture are dragging the category down? One way to find out is to explore the

sub-categories under the Furniture category.

Try the Show Me method again to identify a promising chart type for the columns of interest: Category, Sub-Category, and Sales. Remember you must control-click (Windows) or command-click (Mac) the columns of interest after first revealing the Show Me pane.

- Click on several of the Show Me suggestions to see what they would look like.
- Finally, choose *Horizontal Bars* from Show me and close the Show Me pane.

You know that there is a hierarchical relationship between Category and Sub-Category. Tableau has a technique for expressing that relationship in a way that you can drill down from the more general to the more specific. This technique is to drag and drop columns on top of each other in the data window.

- Drag Sub-Category on top of Category in the Dimensions part of the Data pane. A *Create Hierarchy* dialog box appears. Name the hierarchy *Products* in that dialog box. It now appears in Dimensions with a hierarchy symbol beside it.
- Drag Product Name onto the Products hierarchy.
- Now use the plus and minus on the Category pill to expand and contract the hierarchy.
- You will also find that there are pluses and minuses you can manipulate in the same way inside Sheet 4.
- Make sure that both categories and subcategories are visible when you are done exploring the pluses and minuses.
- Swap the axes using the same Swap Rows and Columns icon on the ribbon you used before.
- Use the Standard button on the ribbon to resize the chart to Entire View.
- Click the Category pill. Then click the Sort Descending button in the ribbon. Now Technology is on the left, fol-

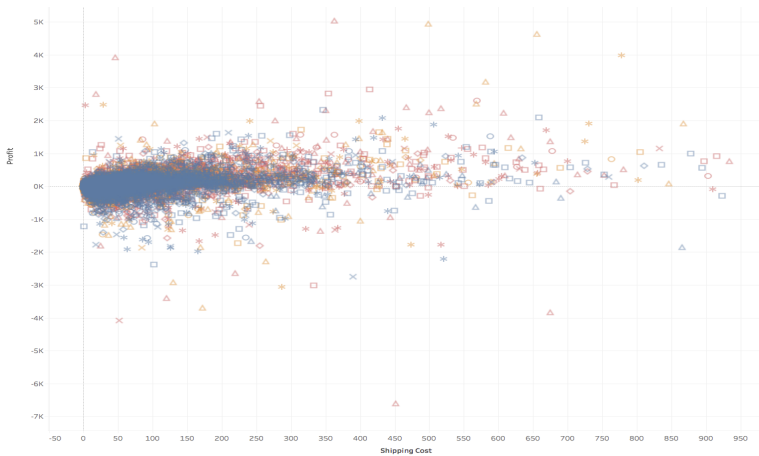
lowed by Office Supplies. Technology has the most total sales, followed by Office Supplies.

- Click on the tiny barchart icon on the left side of the barchart, adjacent to the word Sales. When you click it repeatedly, it changes the sort order of the sub-categories within categories, without changing the categories.
- Choose Show Mark Labels from the menu to display the sales for each bar. Then choose it again to hide them.
- Drag Profit to Color and observe that Tables are a significant profit problem in the Furniture category despite their sales volume.
- Drag Market to the left of Sheet 4. A dashed vertical line should appear and, when you release the mouse, barchart should expand into seven small charts, one for each market. You can see that Table profitability is a problem in four of the seven markets.
- Group some of the office supply columns together. Select the captions for all but the first three by shift-clicking. In the dialog box that appears, select group members.
- Rename the aggregated category by clicking its caption and selecting Edit Alias from the resulting menu. Call it “small office supplies.”
- Simplify the view. Swap the axes, change Fit to Standard, remove Market from the Columns shelf, and hide the field labels for Category and Sub-Category by right clicking on either of them and selecting “Hide Field Labels for Rows.”
- Rename this sheet Sales by Category.

CREATE A SHEET TO EXPLORE SHIPPING COST

Suppose you have a hunch that the low profitability in furniture is related to shipping costs. Now create a new sheet to explore that possibility.

- Select New Worksheet.



- Drag Profit to the Rows shelf.
- Drag Shipping Cost to the Columns shelf.
- Drag Market onto the Shape shelf.
- Drag Category onto the Color shelf.
- Drag Customer ID onto the Detail shelf.

Now Tableau makes a mark for each customer, shaped by Market and colored by Category. (There may actually be more than one mark per customer if a customer has ordered from more than one category or in more than one market.) Each mark represents the sum of profit and shipping cost per customer per market per category.

- Click on the Analysis menu tab at the top of the window. Notice that *aggregate measures* is checked. Uncheck it and now there is a single mark for each transaction. Any transactions below the horizontal 0K line are not profitable!
- On the Marks shelf, click to the left of Customer ID and select Label. Now the Customer ID labels each point. Go back to that same menu and select Detail or click Undo.
- Another way to label is to drag something onto label, for example, drag Sales onto the Label shelf.

- Click the Label shelf and click the three dots next to Text. An edit label dialog appears. Add a dollar sign before the angle brackets surrounding sales. Now the labels will display in dollars.
- Add trend lines. Switch from the data pane on the left to the Analytics pane.
- Select Trend line from the Analytics pane and drag it onto Sheet 5. When it is over Sheet 5 a menu appears. Drag the Trend Line to the box marked Linear. Notice that three trend lines appear, one for each category. Observe that the Furniture line shows that profits rise less with shipping costs than do the profits in the other two categories.
- Look at the mark for one of the low profit / high shipping cost transactions. A tooltip appears.
- Click on the point. A menu appears on the tooltip. Select View Data on the tooltip. A new panel appears with a summary. Click on Full Data and look at the details of the transaction. Close the window.
- Remove Sales from the Marks shelf. The numbers vanish.
- Right click in Sheet 5 and uncheck Show Trend lines.
- Call this sheet Customer Breakdown.

COMPILE A DASHBOARD TO SHARE

A dashboard is the Tableau way to share multiple views of data. You can integrate each of the sheets you've created into a single dashboard to tell a story of data exploration.

- Click on the New Dashboard icon at the bottom of the window.
- Click on the options triangle beside size and choose Laptop Browser, 800 × 600.
- Notice that all the sheets you created are listed on the left and you can hover over them to see a preview.
- Drag Global Sales and Profits into the view.
- Drag Sales by Category into the bottom half of the view.

- Drag Customer Breakdown into the bottom right quadrant of the view.
- In both cases, Sales by Category and Customer Breakdown, make sure Fit is set to Entire View. Right click on the panel, then use the More Options triangle on the tab to find the Fit option.
- Right click on the Dashboard tab and rename it Sales Dashboard.
- Add the name to the dashboard itself by clicking Show dashboard title in the lower left corner.
- Edit the title by double-clicking it, bringing up an edit dialog.
- In the dialog, change the size of the title from 18 to 16 and center it. Click OK to accept your changes.
- Click on the Category menu in the upper right to change the map. Suppose, though, that you want all three visualizations to change to the selected categories?
- Right click on the Category panel and choose Apply to Worksheets from the menu. Choose All Using This Data Source from the submenu. Now all three visualizations change according to the selection.
- Click on the red mark over Auckland, NZ. Suppose you would like to know more about that state. On the map's tab, choose More Options. From the menu, choose Use as Filter. Now you can click on any circle in the map and see the related info in the other two visualizations.

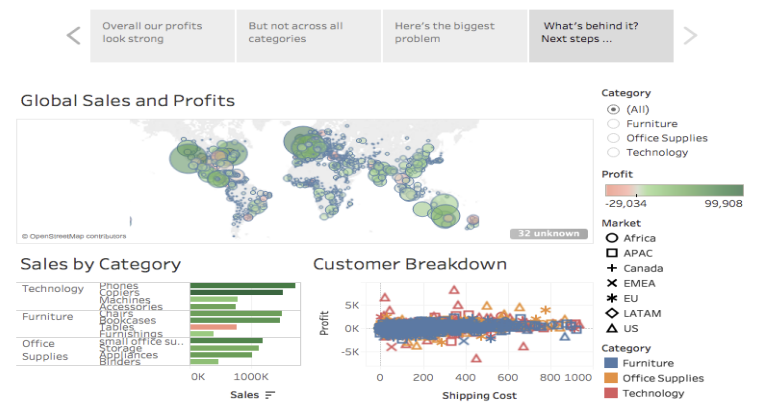
CREATE STORY POINTS

Another way to view the visualizations you have created is through what Tableau calls *story points*. This consists of a navigator bar over a series of visualizations. By clicking on the navigator bar, you step through the story points, capturing each visualization at a specific point.

- Click on the New Story icon.

- Change the size to Laptop Browser 800 × 600.
- Drag Global Sales and Profits onto the view.
- Add a caption, saying “Overall, our profits look strong”.
- Note that the visualization is still interactive. Test that by selecting categories.
- Click on Furniture on the Category panel. A menu appears above the caption. Select Save as New.
- Add a new caption, “But not across all categories”.
- Drag the *Drag to add text* rectangle onto the visualization.
- Add the text “Furniture appears to have poor profitability in some areas” and resize the text to 12.
- Double click on Sales by Category to create a third story point. Add the caption “Here’s the biggest problem”.
- Again add a callout by dragging the *Drag to add text* rectangle to the bottom of the view. Say “Tables are dragging down Furniture’s profits.”
- Drag the Sales Dashboard to the right of the list of captions and add it as a fourth story point.
- Notice that the Sales Dashboard does not fit very well.
- Click on the Go To Sheet icon to the right of the Sales Dashboard’s name in the Story pane.
- Resize it to “Fit to Story”. Also take off the Sales Dashboard title.
- Now return to the Story 1 window and notice that it fits better.
- Add a caption “What’s behind it? Next steps ...”
- Click on Furniture on the Category panel and then select Update from above the caption.
- Double click on Story 1, bringing up an edit dialog.
- Rename it to “Profitability: The Whole Story”.
- Save it as a packaged workbook. Give it a filename of globalSuperstore. Since it is a packaged workbook you will not need the separate spreadsheet file Global Superstore.xls to work with it.

Profitability: The Whole Story



DISCUSSION

Now that you have completed an exercise using Tableau your next step should be to reflect on what you have done. There are a number of discussion questions to consider.

1. The menu structure of Tableau may change drastically from one version to the next. Can you navigate this tutorial in the next (or previous) version of Tableau?
2. Different chart types are recommended for different types of data. How would you characterize these chart types? What is it about the data that leads to the recommendations?
3. Is Tableau biased in any way? Is it easier to make some kinds of arguments with Tableau than others?
4. Tableau accepts live data and allows you to connect your visualization to live, changing data. Suppose the point you are trying to make is no longer supported by data in future. Will you be blamed for that change?

5. Tableau makes a lot of default choices that you changed. Do you think it could (or should) learn your preferences over time? What would it take to do that?
6. What is a data visualization? What would have to be subtracted from these pictures so that they could not be called data visualizations?